

Insurance Claim Timeline Retrieval System

GenAI Multi-Agent Systems — Midterm

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System Overview

This project implements a multi-agent GenAI system for querying an insurance claim timeline. A ManagerAgent routes user queries to a SummarizationAgent for high-level timeline questions, or to a NeedleAgent for precise factual retrieval. The system uses hierarchical RAG indexing with an Auto-Merging Retriever over a synthetic, time-stamped claim dataset, and integrates MCP-style tools for date differences and rental cost computation.

Evaluation Results

Achieved Full Score Across All Quality Metrics

- Correctness: 5.0 / 5.0
- Relevancy: 5.0 / 5.0
- Recall: 5.0 / 5.0

✓ All 8 evaluation queries scored 5/5 across all metrics.

MCP Usage

MCP-style tools are used to extend the LLM beyond pure text reasoning by delegating precise, auditable computations (date differences and rental cost calculations) to explicit tools, while the agents focus on retrieval, routing, and explanation.